

Project Idea Summaries: Inter IIT Tech Meet 14.0

Pathway Problem Statement Innovations

Idea 1: Multi-Agent AI Investment Decision System

Core Idea (from PS)

To build an AI system that acts as an investment team, ingesting live multimodal signals (market APIs, news PDFs, social feeds) to score assets, backtest strategies, and propose trades.

Innovation: Multi-Agent Risk & Conviction

Instead of a single "buy/sell" model, this system uses a multi-agent architecture for a more robust decision:

- **Analyst Agent (GenAI Twist):** A GenAI agent (using a static LLM) reads and understands streaming analyst reports, news, and filings. Its primary job is not just sentiment analysis, but to find the level of **disagreement** or **consensus** among experts, outputting a real-time "Conviction Score."
- **Risk Agent (Pathway Innovation):** This agent leverages Pathway's **stateful windows** on the streaming portfolio data. Its job is to calculate the **overall portfolio volatility** in real-time, rather than just individual asset risk. The portfolio data itself is a real-time join of the system's own "trade log" stream and the "live market price" stream.
- **Investment Agent (The Boss):** The main agent makes the final decision. It combines inputs from all sources:
 1. Live market data (the signal).
 2. The "Conviction Score" (the confidence in the signal).
 3. The "Portfolio Volatility" (the capacity to take on risk).

This allows the system to manage not just *what* to buy, but *how much risk* to take, effectively handling market uncertainty.

Idea 2: Continuous KYC Screener with Deepfake Detection

Core Idea (from PS)

To build an interactive KYC copilot that automates Level-1 checks. It would extract data from uploaded documents, screen names against sanctions lists and adverse media, and use LLMs to summarize hits for a human-in-the-loop.

Innovation: Continuous KYC (cKYC) Agent

The primary innovation is moving KYC from a one-time onboarding check to a **continuous, real-time process**.

- **The Agent's Role:** A "cKYC Agent" uses Pathway's streaming connectors to constantly monitor the entire customer base against live streams of adverse media, sanctions lists, and other risk signals.

- **GenAI Twist (The Analyst):** When a new risk is detected for an *existing* customer (e.g., a customer from 2 years ago appears in a new fraud report), the GenAI agent activates:
 1. It summarizes the new risk.
 2. It pulls the customer's *original* KYC file from the live vector index.
 3. It compares the new risk against the original file's information.
 4. It sends a detailed "Risk Update" alert to the human analyst, providing a clear summary and evidence from both sources.

Add-on: Real-Time Deepfake Detection

This integrates deepfake detection into both onboarding and continuous monitoring:

- **At Onboarding (vKYC):** During the live Video KYC call, the video stream is piped through Pathway to a deterministic deepfake detection model. This model checks for liveness (e.g., eye-blinking, skin texture). If the model has **low confidence**, it triggers the GenAI agent to alert the human operator and suggest specific liveness-proving questions (e.g., "Please raise your right hand").
- **During Continuous Monitoring:** The cKYC Agent also monitors *video* media streams (social media, news sites). If a customer's face is detected in a new video, the deepfake model runs. If the video is flagged as a high-risk deepfake (e.g., showing the customer in a scam), the GenAI analyst alerts the bank to the potential **reputational risk** and **identity fraud**.